



易 灵 思

TinyML

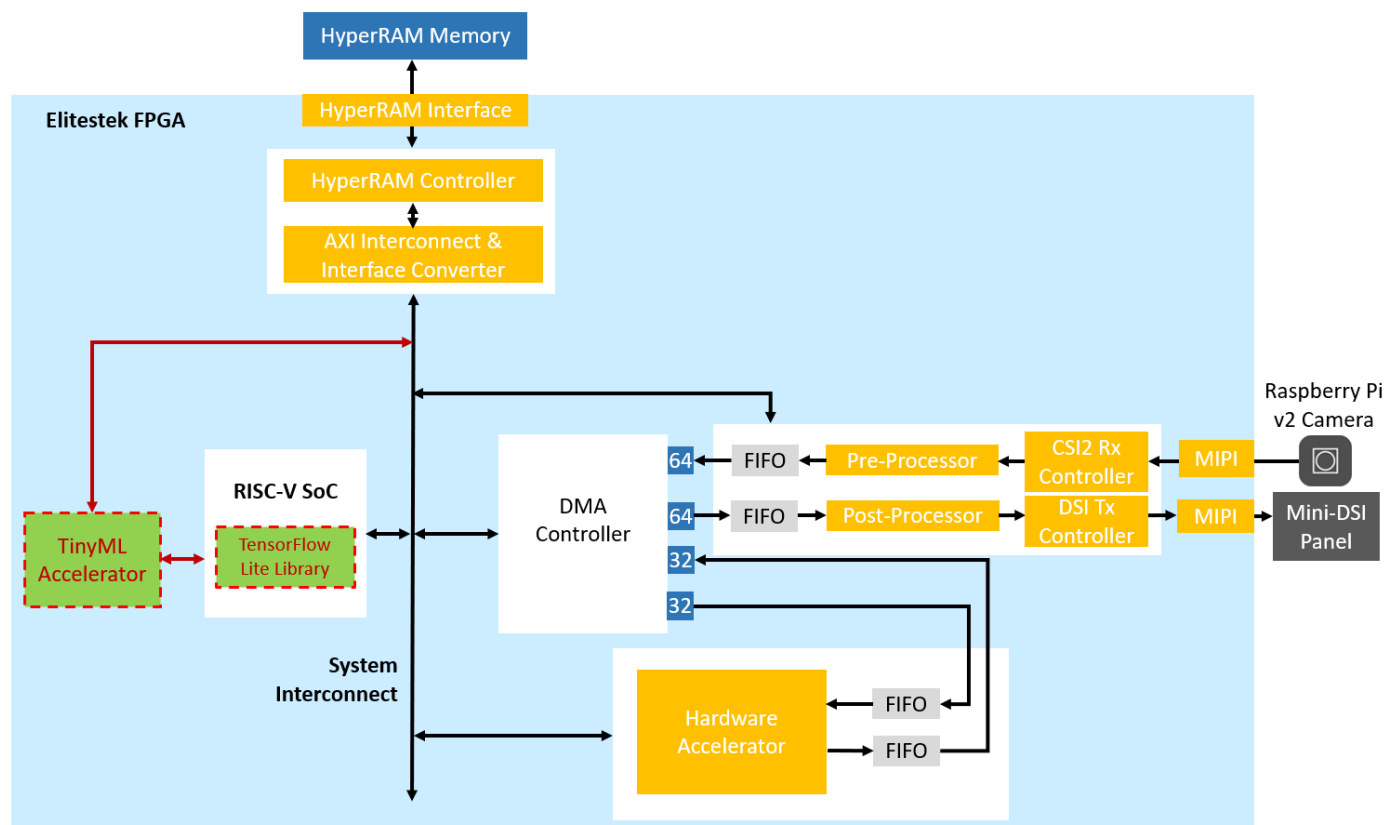
易灵思AI加速平台简介

国产化中高端FPGA领军厂商 —— 加速您的创芯！

易灵思(深圳)科技有限公司

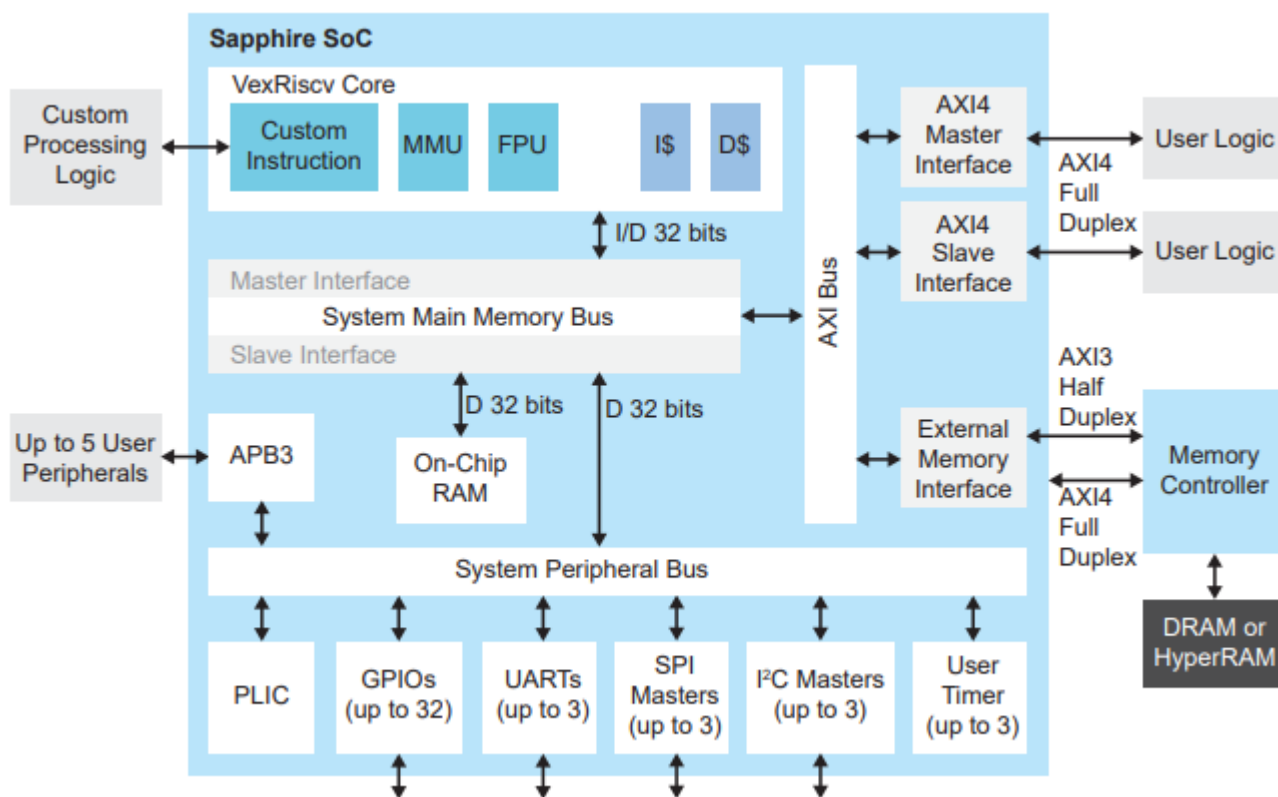
2025.6

Herbert



➤ 基于SapphireSoc的
RISC-V开发

➤ 完整的视频处理流程



➤ 开源RISC-V指令集

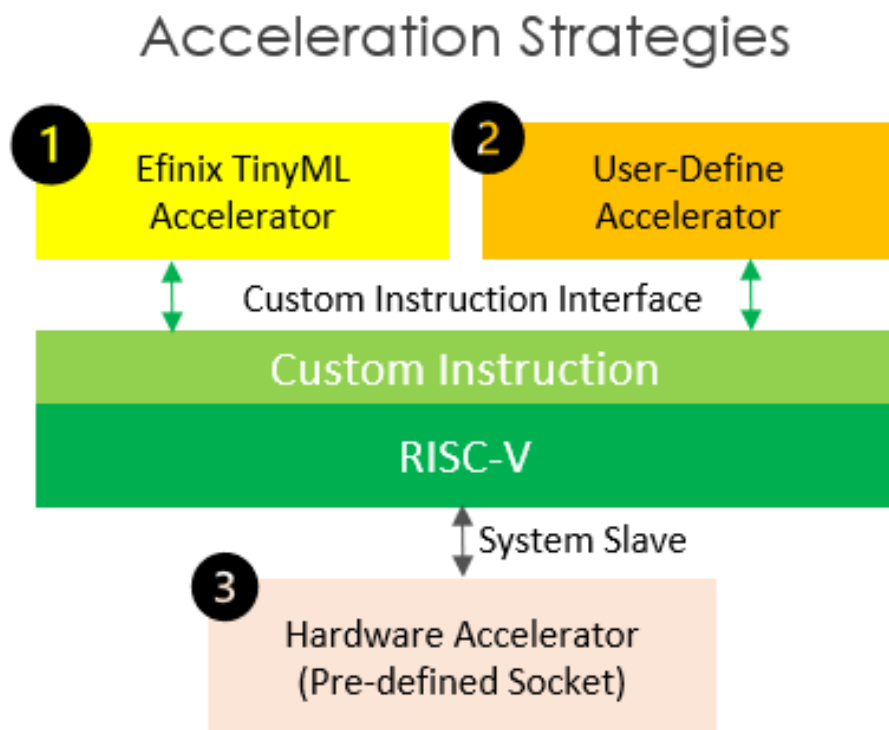
➤ 高可定制性

- 指令
- Cache
- FPU
- 存储
- 外设

➤ 高性能32位实现

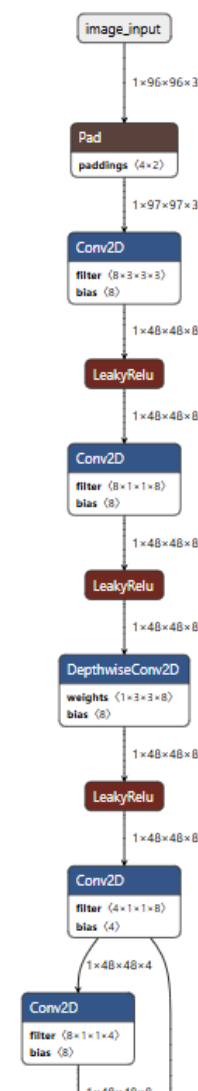
- DMIPS: 1.16
- Fmax: > 350 MHz

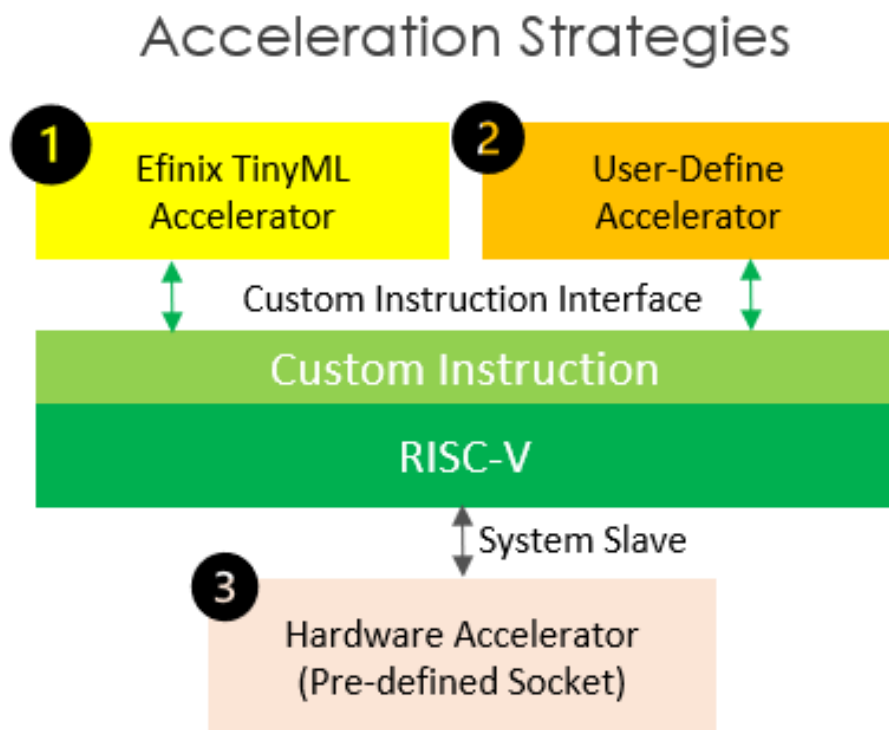
➤ 单核/多核实现



- Tensorflow Lite推理框架
- INT8量化模型加速
- 基于RISCV的软件开发
- 可灵活配置加速模块资源
- 可自定义加速算子

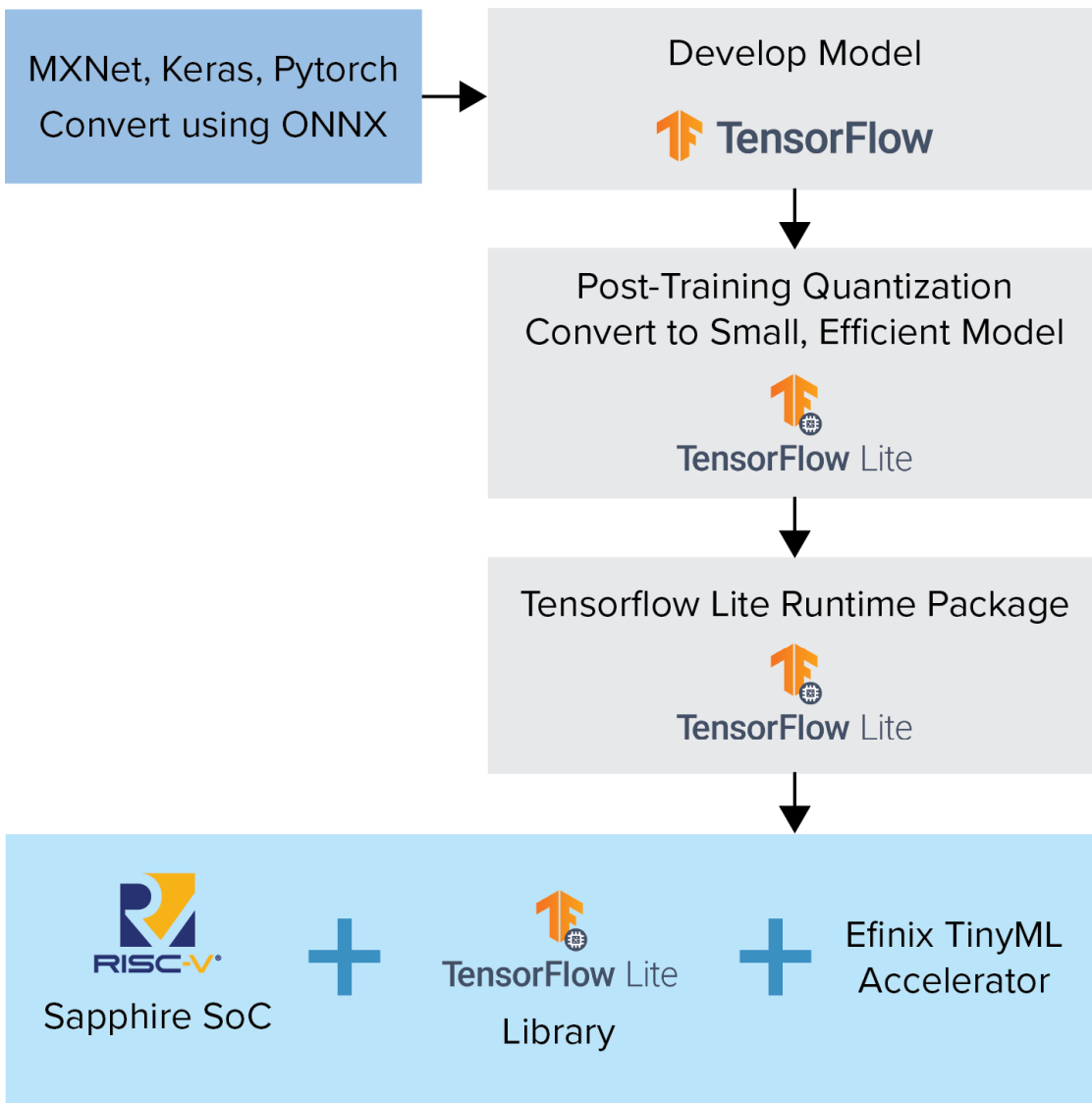
- yolo_person_detect: 用于行人检测
- 主要包含计算的层:
 - Conv2D/Depthwise: 占据了超过90%的计算量
 - 少量算数运算层: Add
 - 激活函数层: LeakyRelu



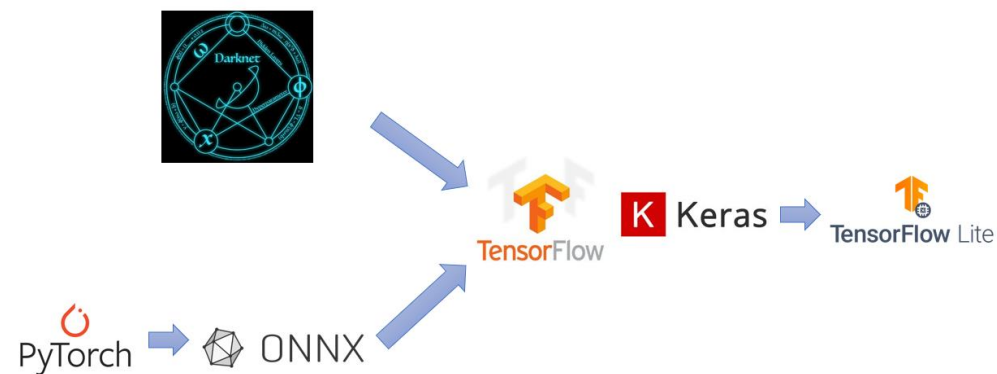


- 卷积: Conv2D、Depthwise
- 算数: Add/Mul
- 全连接: FullyConnected
- 激活函数: Relu/LeaklyRelu

TinyML使用流程



- 模型开发
- 模型量化
- 数据导出
- 硬件部署



Elitestek TinyML Generator

Parameter	Value
SYSTEM	
AXI_DW	128
CONV_DEPTHW_MODE	STANDARD
CONV_DEPTHW_STD_IN_PARALLEL	8
CONV_DEPTHW_STD_OUT_PARALLEL	4
CONV_DEPTHW_LITE_PARALLEL	4
ADD_MODE	STANDARD
FC_MODE	STANDARD
MUL_MODE	DISABLE
MIN_MAX_MODE	DISABLE
TINYML_CACHE	ENABLE
CACHE_DEPTH	512

Resource Estimator

Layer/Module	LUTs	FFs	ADDs
1 CONV_DEPTHW	12516	8870	5323
2 ADD	1904	2674	747
3 FC	1842	2351	976
4 TINYML_CACHE	880	488	172
5 COMMON	2873	1073	116
6 TINYML_ACCELERATOR	20015	15456	7334

NOTES:

1. Efinix TinyML platform supports .tflite model with full integer quantization.
2. LITE accelerator is a lightweight accelerator with less resource usage.
3. STANDARD accelerator is a high performance accelerator with more resource usage.
4. TinyML cache is an optional feature to speed-up data access of STANDARD mode accelerator.
5. Resource Estimator estimates resource usage of TinyML accelerator based on configured parameter setting for Efinix Titanium FPGAs. Open .tflite model file to get started.

Model File: D:/Elitestek/TinyML/TinyML_0603/tools/tinymml_generator/model/resnet_image_classify.tflite

Open Generate

- 自动分析加速算子
- 加速器参数配置
- 资源占用预估
- 导出数据文件

➤ TinyML IP、外设、系统集成

Efinity工程

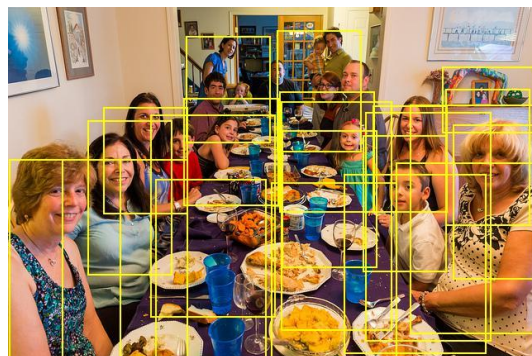
TinyML软件

➤ TinyML模型加载、推理、结果输出

➤ 全工程烧录并运行

FPGA设备

Application	Framework	Model format	Model	Input Size	Dataset	Quality Target (%)
Yolo Person Detection	Darknet	.cfg, .weights	Yolo	96x96x3	COCO(person)	20.30 (mAP@0.5)
Person Detection	Tensorflow	.pb	MobilenetV1	96x96x1	Visual Wake Words	84.0
Image Classification	Tensorflow	.h5	ResNet	32x32x3	CIFAR10	85.0
Keyword Spotting	Tensorflow	.pb	DS-CNN	49x10x1	Speech Commands	90.0
Face Landmark	Tensorflow	.pb	MediaPipe	192x192x3	Charade	468(3D Landmark)
Anomaly Detection	Tensorflow	.h5	AutoEncoder	1x640	ToyADMOS	0.85 (AUC)



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